

PHYSICS

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7. (2) a convex mirror.

8. (3).

9. (2) There is no atmosphere at great heights.

10. (2) Refraction, dispersion and total internal reflection.

11. (3) low resistance and high melting point.

12. (3)

13. (3) The pattern of Magnetic field associated with the solenoid is different from the pattern of the magnetic field around a bar magnet.

23. The conductor will not experience any force if it ~~is kept~~ the angle between the current and the magnetic field is 0° or 180° , i.e., parallel.

24. The image formed by a plane mirror will be of the same size as the object, virtual, erect & laterally inverted.

25. we know,

$$\omega = \frac{V}{r} \quad V = \omega r$$

$$V = \omega r$$

$$\omega = \frac{V}{r} \rightarrow (1)$$

$$I = \frac{q}{t}$$

$$q = It \rightarrow (2)$$

Substituting (2) in (1), we get.

$$W = V \times It$$

$$W = VIt \rightarrow (3)$$

We know, by Ohm's law

$$V = IR \rightarrow (4)$$

Substituting (4) in (3), we get

$$H = IR \times It$$

$$\boxed{H = I^2 R t} \quad [H \text{ is the conversion of electrical energy to Heat energy}]$$

$$l = 12 \text{ m}; A = \pi \times 2 \times 10^{-4} \times 2 \times 10^{-4} = \pi \times 4 \times 10^{-8}$$

$$\rho = 3.14 \times 10^{-8} \Omega \text{ m}.$$

$$l = \frac{RA}{A} \Rightarrow R = \frac{\rho l}{A} \Rightarrow R = \frac{3.14 \times 10^{-8} \times 12}{3.14 \times 4 \times 10^{-8}} = 3 \Omega.$$

$$\boxed{R = 3 \Omega}$$

Q1.(i) - 0.5 D indicates the usage of concave lens which is used for correction of myopia.
 $P = \frac{1}{f} \Rightarrow f = \frac{1}{P} \Rightarrow f = \frac{1}{-0.5} \Rightarrow f = -\frac{1}{0.5} \Rightarrow \boxed{f = -2 \text{ m}}.$

the focal length is 2m.

(ii) +1.5 D indicates the usage of convex lens which is used for correction of hypermetropia.

$$f = \frac{1}{P} \Rightarrow f = \frac{1}{1.5} \Rightarrow f = \frac{10}{15} \Rightarrow f = \frac{2}{3} \Rightarrow \boxed{f = 0.67 \text{ m}}.$$

the focal length is 0.67m.

35. (i) The brightness of the bulbs B_2 & B_3 increases.

(ii) A_1 & A_3 ammeters show a reading of 1.5 A each, A_2 shows a reading of 0 A. Ammeter A shows the same reading of 3 A.

(iii) $P = I^2 R$ / $P = VI$; $V = 4.5V$; $I = 3A$

$$P = \frac{4.5}{10} \times 3 = \frac{13.5}{10} \Rightarrow \boxed{P = 13.5W}$$

Power dissipated is 13.5 W. when all 3 bulbs glow in the circuit

38. (a) ~~400nm-700nm~~ 380nm-780nm.

(b) Because the refractive index of the prism is different for different colours of light.

(b)

38. (a) 380-780nm.

(b) In this, the light refracts, meaning light is broken or splits up to its frequencies or wavelengths.

(c) The refractive index of the glass prism is different for different colours of light.